

**D. Y. PATIL COLLEGE OF ENGINEERING & TECHNOLOGY**  
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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**



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**B.TECH. PROJECT SYNOPSIS**

*on*

**StartupSync : An AI-Driven Integrated Startup Ecosystem**

**Submitted By:**

**Group ID : G40**

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## 1. Introduction

Entrepreneurship has emerged as one of the major drivers of economic development, innovation, and employment generation.[\[6\]](#) While there are numerous government initiatives, incubators, investors, and platforms that support entrepreneurship, these resources are siloed and fragmented, making it challenging for entrepreneurs to navigate and leverage them effectively. Most existing platforms offer either job recruitment, funding opportunities, or mentorship but fail to provide an integrated view of the entire ecosystem.[\[2\]](#)

The proposed project- “StartupSync”, envisions developing an AI-driven integrated startup management platform, addressing the challenges by offering a unified interface for connecting entrepreneurs, mentors, investors, interns, incubators, and service providers. The system is designed to utilize intelligent recommendation approaches for personalized startup opportunities, internships, mentorship, funding schemes, and incubation support. [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)

The proposed work aligns well with the current surge in artificial intelligence, recommendation systems, digital entrepreneurship, and startup ecosystems, making the project industrially relevant and research-worthy.[\[1\]](#) [\[2\]](#) [\[5\]](#)

## 2. Literature Survey and Need of the Work

### Literature Survey

| Sr. No. | Title                                                                                                                   | Author(s) & Year                                           | Summary                                                                                                  | Limitations                                                                     |
|---------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1       | A Data-Driven Hybrid Recommendation Framework for Startup Investment Platforms                                          | Sonia Setia et al. (2025)                                  | Combines multiple recommendation techniques for personalized startup investment recommendations.         | Depends on data quality and faces cold-start and scalability challenges.        |
| 2       | Advancing Startup Ecosystems Through AI-Driven Matchmaking                                                              | Ninda Lutfiani et al. (2025)                               | Explains collaborative and content-based filtering.                                                      | No real-time personalized matchmaking system.                                   |
| 3       | Startup-Investor Connect using ML                                                                                       | P. Niharika et al. (2026)                                  | Uses ML to match startups with suitable investors based on profile compatibility.                        | Does not consider social or behavioral factors.                                 |
| 4       | Content-Based Recommendation Engine Using Term Frequency-Inverse Document Frequency Vectorization and Cosine Similarity | I. Lumintu (2023)                                          | Uses TF-IDF Vectorization and Cosine Similarity to develop a content-based recommendation engine.        | Directly supports the proposed Content-Based Filtering approach in StartupSync. |
| 5       | Introducing a Predictive-Oriented Multi-Stakeholder Recommender System Model                                            | Reza Davoudabadi, Ahmad Makui & Mohammad R. Rasouli (2026) | Recommender system that balances user and organizational preferences using flexibility, Prospect Theory. | Evaluation is based on a simulated dataset of only 100 users and 30 products.   |
| 6       | Predicting the Success of Startups Using a Machine Learning Approach                                                    | Mona Razaghzadeh Bidgoli et al. (2024)                     | Uses ML classification and clustering to predict startup success.                                        | Limited to 400 startups.                                                        |

## Need of the Work

From the above literature survey, it is evident that the current startup management platforms are fragmented, offering disparate services without personalized recommendations.[\[2\]](#) Entrepreneurs have to explore various portals to identify internships, funding, mentors, government subsidies, legal help, and other resources. Similarly, investors and mentors have to go through several listings to identify the most promising startups.[\[3\]](#) [\[4\]](#) The proposed work, StartupSync, aims to provide an integrated platform for managing startups with AI-driven recommendation systems for personalized internships, funding, mentors, and investors.[\[1\]](#) [\[2\]](#) [\[5\]](#)

### 3. Problem Statement

The current startup management platforms are disjointed and offer limited personalized assistance. Entrepreneurs find it challenging to identify the most suitable mentors, investors, internships, government schemes, and technical support. Investors, on the other hand, have to sift through various proposals before identifying scalable startups,[\[3\]](#) [\[4\]](#) and students are unable to find startup internships.

The proposed work aims to design and develop an AI-driven integrated startup management platform to address these challenges.[\[1\]](#) [\[2\]](#)

## 4. Objectives

### Primary Objectives

1. To design and develop a centralized platform for managing startup needs such as funding, internships, mentors, government schemes, and technical support.
2. To employ artificial intelligence and machine learning techniques to provide personalized recommendations to entrepreneurs, investors, and mentors.
3. To enhance collaboration between startups, investors, mentors, students, and service providers.
4. To facilitate improved resource discovery and engagement for better startup growth.

## 5. Proposed Work

### 5.1 System Architecture

The proposed system will follow a Three-Tier MERN Stack Architecture, ensuring scalability, modularity, security, and maintainability. The architecture will consist of the following layers:

#### 1. Presentation Layer (Frontend)

The frontend will be built using React.js, HTML, CSS, and JavaScript to render responsive web pages. The system will provide a user-friendly interface through which entrepreneurs, investors, mentors, students, incubators, and administrators will securely access and manage information. The various user roles will interact with the system through dashboards that will offer tailored insights.[\[15\]](#)

#### 2. Application Layer (Backend)

The application layer will comprise Node.js and Express.js, which will provide RESTful APIs for handling requests and queries from the frontend. It will manage user authentication, authorization, and business logic, and will communicate with the database to retrieve the necessary information.[\[8\]](#) [\[9\]](#)

#### 3. Data Layer (Database)

The database will be built using MongoDB and will store user profiles, startup details, internship information, mentor details, funding proposals, government schemes, notifications, and recommendation data.

#### AI Recommendation Engine

The AI recommendation engine can be implemented as a separate component that will recommend mentors, investors, internships, and government schemes based on the user's interests, skills, domain, and past interactions.[\[2\]](#) [\[3\]](#) [\[4\]](#) Initially, a

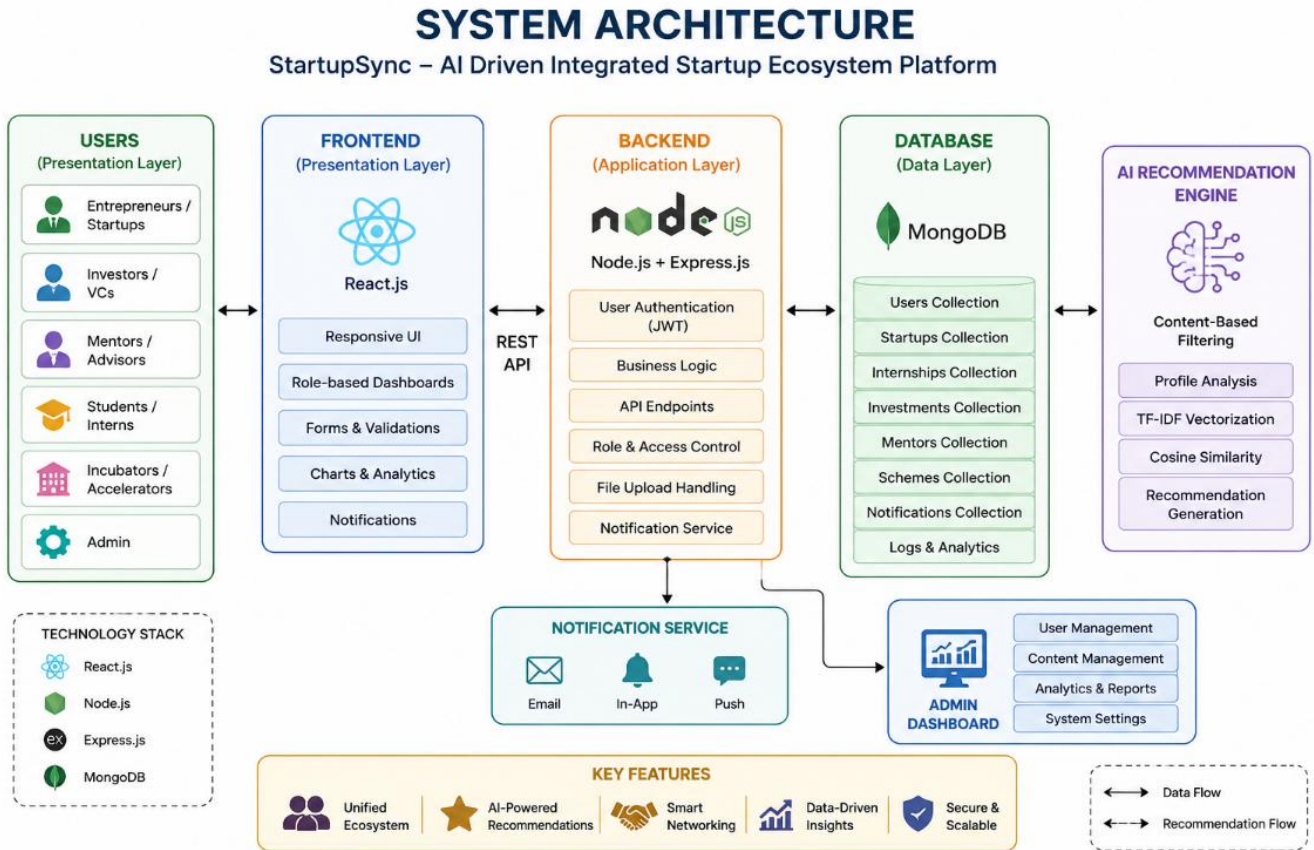
Content-Based Filtering approach using TF-IDF Vectorization and Cosine Similarity can be used to recommend similar items.[\[7\]](#) [\[12\]](#) [\[13\]](#) More advanced techniques, such as Collaborative Filtering, may be integrated later.[\[1\]](#) [\[5\]](#)

### **Overall Workflow**

1. The user will interact with the React.js frontend.
2. The frontend will send requests to the Express.js REST APIs.
3. The backend will handle the requests and query the MongoDB database.
4. The Recommendation Engine will recommend items based on similar domains and skills.
5. The results will be sent back to the frontend and rendered to the user.

### **Architecture Components:**

- React.js Frontend
- Node.js & Express.js Backend
- MongoDB Database
- JWT Authentication
- AI Recommendation Engine
- Notification Service
- Admin Dashboard
- REST API Communication



## 5.2 Mechanism and Approach

The proposed mechanism makes use of the MERN Stack (MongoDB, Express.js, React.js, Node.js)

- React.js offers a responsive frontend, whereas Node.js and Express.js provide the business logic and database connectivity.[\[8\]](#) [\[9\]](#) [\[15\]](#)
- MongoDB stores user data and startup-related information.[\[10\]](#)
- The system uses JWT Authentication for secure user verification.
- Additionally, the system uses an AI recommendation engine to recommend similar items based on the user's interests, domain, and previously viewed items.[\[1\]](#) [\[2\]](#) [\[3\]](#) Initially, we can make use of Content-Based Filtering (TF-IDF, Cosine Similarity)[\[7\]](#) [\[12\]](#) [\[13\]](#) and later adopt more advanced recommendation techniques such as Collaborative Filtering.[\[1\]](#) [\[5\]](#)

## 5.3 Project Modules

### Module 1 – User Registration & Authentication

This module will allow users such as entrepreneurs, investors, mentors, students, incubators, and administrators to register and securely log in to the system.

#### Technology:

- React.js – frontend registration and login interfaces [\[15\]](#)
- Node.js – server-side application runtime [\[8\]](#)
- Express.js – REST API and authentication request handling [\[9\]](#)
- MongoDB – storage of user profiles and account information [\[10\]](#)
- Mongoose – schema-based modeling and validation of MongoDB data [\[11\]](#)

### Module 2 – Startup Management

This module will allow entrepreneurs and startup representatives to create and manage startup profiles. Users will be able to maintain information such as startup name, industry/domain, business stage, funding requirements, team information, skills, location, and other relevant details. [\[3\]](#) [\[6\]](#)

#### Technology:

- React.js – startup profile creation and management UI [\[15\]](#)
- Node.js + Express.js – startup management APIs [\[8\]](#) [\[9\]](#)
- MongoDB + Mongoose – storage and management of startup profiles [\[10\]](#) [\[11\]](#)

### Module 3 – Investor & Mentor Portal

This module will provide dedicated portals for investors and mentors. Investors will be able to create profiles containing investment interests, preferred sectors, investment ranges, and funding stages. Startups will then be able to discover suitable investors and mentors through profile-based matching and AI recommendations. [\[2\]](#) [\[3\]](#) [\[4\]](#)

#### Technology:

- React.js – investor and mentor dashboards [\[15\]](#)
- Node.js + Express.js – portal APIs and profile management [\[8\]](#) [\[9\]](#)
- MongoDB + Mongoose – investor/mentor profile storage [\[10\]](#) [\[11\]](#)
- REST API – communication between frontend and backend [\[10\]](#)

### Module 4 – Internship & Recruitment Module

This module will provide a platform for startups and organizations to publish internship and recruitment opportunities. Students and other eligible users will be able to browse available opportunities, view eligibility requirements, apply for positions, and track application status.

#### Technology:

- React.js – internship listing, search, and application interfaces [\[15\]](#)
- Node.js + Express.js – internship and recruitment APIs [\[8\]](#) [\[9\]](#)
- MongoDB + Mongoose – internship and application data [\[10\]](#) [\[11\]](#)
- TF-IDF and Cosine Similarity – matching users with relevant opportunities [\[12\]](#) [\[13\]](#)

## Module 5 – AI Recommendation Engine

This module will act as the core intelligent component of StartupSync. It will analyze user interests, skills, startup domain, funding requirements, preferences, and previous interactions to recommend suitable mentors, investors, internships, and government schemes.[\[2\]](#) [\[3\]](#) [\[4\]](#) Initially, a Content-Based Filtering approach can be implemented using TF-IDF Vectorization and Cosine Similarity.[\[7\]](#) [\[12\]](#) [\[13\]](#) The system may later be extended using collaborative or hybrid recommendation techniques.[\[1\]](#) [\[5\]](#)

### Technology:

- Python – recommendation-engine implementation
- Scikit-learn – machine-learning and text-processing implementation [\[12\]](#) [\[13\]](#)
- TF-IDF Vectorization – conversion of textual profiles/descriptions into numerical feature vectors [\[12\]](#)
- Cosine Similarity – calculation of similarity between user profiles and recommendation items [\[13\]](#)
- Node.js + Express.js – communication between the MERN application and recommendation engine [\[8\]](#) [\[9\]](#)

## 6. Requirements

### 1. Hardware Requirements

#### Hardware

- Intel Core i5 Processor or above
- 8 GB RAM
- 256 GB SSD
- Internet Connection

### 2. Software Requirements

#### Software

- React.js
- Node.js
- Express.js
- MongoDB
- VS Code
- Git & GitHub
- Python (Recommendation Engine)
- TensorFlow / Scikit-Learn (Future Enhancement)

### 3. Specific Devices / Tools / Platforms

#### Tools / Platforms

- MongoDB Atlas
- GitHub
- Figma
- Render / Vercel
- Cloudinary

#### 4. Cost of Implementation

| ITEM              | ESTIMATED COST    |
|-------------------|-------------------|
| Software Tools    | Free              |
| GitHub            | Free Student      |
| MongoDB Atlas     | Free Tier         |
| Render Deployment | Free              |
| Domain            | ₹800 - ₹2000 year |

Approximate Total Cost : **₹1000 – ₹3000**

## 7. Project Plan

| Sr. No. | Month     | Plan                                                |
|---------|-----------|-----------------------------------------------------|
| 1       | July      | Requirement Analysis, Literature Survey             |
| 2       | August    | UI/UX Design, Database Design                       |
| 3       | September | Backend Development (Node.js, Express, MongoDB)     |
| 4       | October   | Frontend Development & Recommendation System        |
| 5       | November  | Testing, Deployment, Documentation & Research Paper |

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## 9. Synopsis Approval Sheet

**Group Number:** [G40]

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**Place:** Kolhapur

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**Sign of Guide**

**Sign of Project Coordinator**

**Sign of HOD**